Quantitative investment and stock price crash risk in China: Perspective of quantitative mutual funds holdings

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ABSTRACT

This study examines the impact of quantitative investment on stock price crash risk from the perspective of quantitative mutual funds holdings. The results show that quantitative mutual funds holdings can significantly reduce stock price crash risk, and this effect is more pronounced in subsamples characterized by executives with overseas backgrounds, higher internal governance efficiency, greater analyst attention, and higher profit volatility. Further research finds that quantitative mutual funds holdings can suppress the risk of stock price crash by smoothing the volatility of stock returns and optimizing the valuation of firms. This study sheds light on the effects of quantitative investment on stock price crash risk.

Keywords: Quantitative investment; Quantitative mutual funds; Stock price crash risk; China

1. Introduction

Quantitative investment refers to a trading strategy that employs quantitative methodologies and computer algorithms to develop trading models and execute buying and selling orders, aiming to achieve consistent and stable returns. In recent years, the capital market has witnessed a rapid accumulation of various types of data due to the rapid development of digital technology. To cope with the complex, high-noise, and non-linear nature of market data, artificial intelligence algorithms like machine learning have been implemented which has strongly driven the advancement of quantitative investment, especially in the domain of quantitative mutual funds. Quantitative investment is deemed more rational and efficient when compared to discretionary investment. Consequently, the widespread implementation of quantitative investment technology is expected to influence investors' investment philosophy, subsequently impacting stock price dynamics.

Despite the substantial literature on quantitative investment research, the primary focus of these studies lies in building more effective quantitative trading strategies (Fang et al., 2022) or determining whether the introduction of new algorithms can improve the profitability and prediction

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accuracy of such strategies (Wang et al., 2022). There is little literature examining the impact of the widespread application of quantitative investment techniques on stock price stability. Quantitative investment, as a highly specialized investment strategy, demands not only proficiency in financial, statistical, and programming knowledge from investors, but also relies heavily on extensive data resources and sophisticated computer hardware. The combination of a high entry barrier and intricate technical demands poses significant challenges for the majority of retail investors in engaging with this field. However, with a low entry threshold for investment amounts, quantitative mutual funds, as a professional investment institution highly regarded by investors, offer retail investors an excellent opportunity to participate in quantitative investment practices. That is why this paper wants to investigate the impact of quantitative investment on stock price crash risk from the perspective of quantitative mutual funds holdings.

According to the theory of behavioral finance, some cognitive biases, such as herding bias (Li and Jiang, 2022; Deng et al., 2018) and over-confidence bias (Kim et al., 2016), contribute significantly to stock price crash. Discretionary investors are susceptible to the influence of instantaneous emotions, which often leads to irrational decision-making. On the contrary, quantitative trading strategies, as specific computer programs developed by learning historical data and summarizing trading experience, can firmly implement the established trading intention and make scientific trading decisions to overcome the interference of cognitive bias on investment. Therefore, the extensive application of quantitative investment technology can effectively contribute to stock price stabilization.

The empirical findings show that an increase in the shareholding ratio of quantitative mutual funds is associated with a decrease in the risk of stock price crash. Moreover, this inhibitory effect is more pronounced in subsamples characterized by executives with overseas backgrounds, higher internal governance efficiency, greater analyst attention, and higher profit volatility. Further research finds that quantitative mutual funds holdings can suppress the risk of stock price crash by smoothing the volatility of stock returns and optimizing the valuation of enterprises.

The main contribution of this study is threefold. First, this paper expands the research of quantitative investment. The existing literature primarily focuses on investigating new algorithms and models' effectiveness in predicting stock returns (Hu et al., 2023), as well as how to apply documented indicators to build more profitable quantitative trading strategies (Fang et al., 2022), disregarding the influence of quantitative investment on the capital market's dynamics. Second, this is the first time for the academic community to investigate the impact of quantitative mutual funds on the capital market. Third, our study enriches the relevant literature on stock price crash risk. It reveals the significance of quantitative investment in fostering the growth of the capital market and ensuring the stability of stock prices, thereby providing valuable evidence supporting the widespread application of quantitative investment and financial technology in the investment industry.

2. Prior literature and hypotheses development

Some scholars attempt to explain the risk of stock price crash based on the theory of bad news hoarding. Given the market's limited tolerance for negative news, managers tend to hide or delay the disclosure of negative information for the motivation of maintaining career prospects and seeking promotion (Graham et al., 2005; Piotroski et al., 2015; Chang et al., 2017). When negative

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news erupts in great quantities, the stock price experiences a collapse. For example, Kong et al. (2023) found that the uncertain news related to COVID-19 exacerbated the risk of stock price crash. Some scholars also try to use behavioral finance theory to attribute stock price crash. For instance, the study conducted by Kim et al. (2016) revealed a positive correlation between overconfident CEOs and the risk of stock price crash. Deng et al. (2018) found that mutual funds herding amplifies stock price crash risk. Jang and Kang (2019) suggest that overpricing driven by institutional investment will lead to stock price crash.

Quantitative investment has been explored in academia and industry for many years, from early multi-factor models to machine learning algorithms that are now widely ongoing discussion. Due to the high dimensionality and low signal-to-noise ratio of stock data, traditional quantitative trading models are becoming increasingly difficult to generate excess returns. As a result, the investment community is eager for the profits that come with cutting-edge technological innovation (Fischer and Krauss, 2018). Simultaneously, certain research findings also provide empirical support for the application of the new algorithm in quantitative investment. For example, Gu et al. (2020) showing that machine learning methods perform observably better than traditional linear regression models. Hu et al. (2023) found that machine learning yields favorable investment returns in simulated quantitative trading of China's A-share market. Therefore, an increasing number of cutting-edge techniques are being employed in quantitative investment.

Quantitative investment possesses two primary advantages in comparison to discretionary investment. Firstly, quantitative investment can collect information more extensively and process data more efficiently. Different from the limited computing power and attention of the human brain, the computing power of quantitative investment strategies is approaching infinity with the continuous improvement of computer hardware and the rapid development of digital technology. This enables it to apply big data technology to collect multi-level information from a broader dimension through the internet. Furthermore, by utilizing advanced information processing technology, quantitative investment has the capability to convert unstructured data such as images, audios, and videos into structured data and conduct real-time monitoring of market information, thereby enhancing investment efficiency and accuracy. Typically, information lag or asymmetry serves as a significant factor contributing to fluctuations in stock prices. The decline in investors' ability to access information online will increase the risk of stock price crash (Xu et al., 2021). By leveraging its powerful capabilities in information collection and processing, quantitative investment can facilitate the efficient and timely integration of heterogeneous information into stock prices, thereby reducing stock price volatility. Based on this analysis, we argue that quantitative investment can reduce the risk of stock price crash by smoothing stock price fluctuation.

Secondly, quantitative investment can overcome cognitive biases and irrational behaviors in discretionary investment. Given the limited rationality of human beings and their susceptibility to various cognitive biases (Linnainmaa et al., 2021; Bazley et al., 2021), it is susceptible for them to make irrational investment decisions when influenced by specific situations or information flows. For example, Lwin et al. (2020) found that investor sentiment in the stock market can be easily affected by different news of COVID-19. Especially when investors are driven by over-optimistic sentiment, they often tend to propel stock prices to surge significantly in a short period of time, resulting in stock prices being severely overestimated. However, once the adverse information concealed by the management is revealed, investors' sentiment is likely to quickly shift from over-optimism to over-pessimism, triggering the burst of the stock price bubble and ultimately leading to

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a crash. By contrast, quantitative investment strategies, which are specific computer programs developed by incorporating previous investment experience, have the ability to make optimal trading decisions that align with the risk-return targets of the strategies in complex information environments, without being influenced by investors' subjective preferences. For instance, Chen and Ren (2022) provide evidence that AI-powered funds can overcome some of the prevalent cognitive biases such as disposition effect and rank effect. Moreover, quantitative investment strategies have the potential to continue growing and evolving, especially with the introduction of cutting-edge algorithms such as machine learning. Quantitative investment, leveraging its objective and rational investment style coupled with the precise analysis of cutting-edge algorithms, can provide a more accurate valuation for stocks, thus effectively preventing the emergence of speculative bubbles. Based on this analysis, it is reasonable to believe that quantitative investment can mitigate the risk of stock price crash by optimizing the valuation of firms.

3. Data and variables

3.1. Independent variable

The key independent variable quantitative mutual funds holdings (*QFHPF*) is constructed through the following steps. First, select the mutual funds that contain one or more of the following keywords in their investment strategy text descriptions: “Quantitative Investment”, “Quantitative Indicators”, “Quantitative Tools”, “Quantitative Model”, “Quantitative Hedge”, “Quantitative Strategy”, “Quantitative Team”, “Quantitative Multifactor”, “Quantitative Timing”, “Quantitative Stock Selection”, “Quantitative Trading”, “Quantitative Method”, “Quantitative Analysis”, “Quantitative Processing”, “Quantitative Screening”, “Quantitative Risk”, “Quantitative Modeling”, “Quantitative Multi-Strategy”, “Quantitative Pricing”, “Quantitative Valuation”, “Algorithm”, “Statistical Arbitrage”, “Backtest”, “Market Neutrality”, “Event-Driven”, “Artificial Intelligence”, “Machine Learning”, “Neural Network”, “Deep Learning”, “Big Data”, “Natural Language Processing”. Second, manually exclude mutual funds that invest in the artificial intelligence technology industry chain rather than utilizing artificial intelligence technology for investment, and categorize these screened mutual funds as quantitative mutual funds. Third, according to the listed company's funds holdings data, calculate the total number of shares held by all mutual funds and quantitative mutual funds at the end of each year during the sample period. Given the potential impact of institutional investors' behavior on the likelihood of stock price crash (Deng et al., 2018; Jang and Kang, 2019), the proxy variable for quantitative mutual funds holdings in this paper is defined as the ratio of shares held by quantitative mutual funds to the total shares held by all mutual funds.

3.2. Dependent variable and control variables

Following the literature (Hutton et al., 2009; Kim et al., 2011; Zuo et al., 2022), this study uses the negative return skewness coefficient (*NCSKEW*) and down-to-up volatility of returns (*DUVOL*) to measure stock price crash risk. The calculation process of *NCSKEW* is shown in Eq. (1), (2) and (3). Where $r_{i,j}$ is the return of stock *i* in week *j*, and $r_{m,j}$ is the value-weighted average return of A-share market in week *j*. Considering asynchronous trading, $r_{m,j-2}$ and $r_{m,j-1}$ denote the lagged

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terms, $r_{m,j+1}$ and $r_{m,j+2}$ denote the leading terms. Additionally, $w_{i,j}$ denotes the firm-specific weekly return of stock *i* in week *j*, and $\varepsilon_{i,j}$ is the residual term of Eq. (1). In Eq. (3), *n* represents the number of effective trading weeks for stock *i* in year *t*. The calculation process of *DUVOL* is shown in Eq. (1), (2) and (4). In addition to the variables mentioned above, the calculation of *DUVOL* requires grouping based on whether the firm-specific weekly return is above the annual mean. Weeks with returns above the annual mean are designated as up weeks, while weeks with returns below the annual mean are designated as down weeks. In Eq. (4), n_u and n_d represent the number of up weeks and down weeks for stock *i* in year *t*, respectively. Larger values of *NCSKEW* and *DUVOL* imply higher stock price crash risk.

$$r_{i,j} = \alpha + \beta_1 r_{m,j-2} + \beta_2 r_{m,j-1} + \beta_3 r_{m,j} + \beta_4 r_{m,j+1} + \beta_5 r_{m,j+2} + \varepsilon_{i,j} \quad (1)$$

$$w_{i,j} = \ln(1 + \varepsilon_{i,j}) \quad (2)$$

$$NCSKEW_{i,t} = -\frac{n(n-1)^{\frac{3}{2}} \sum w_{i,j}^3}{(n-1)(n-2)\left(\sum w_{i,j}^2\right)^{\frac{3}{2}}} \quad (3)$$

$$DUVOL_{i,t} = ln\left[\frac{n_u - 1}{n_d - 1} \frac{\sum_{Down} w_{i,j}^2}{\sum_{Up} w_{i,j}^2}\right] \quad (4)$$

Control variables include: change in turnover rate (*Dturn*), firm age (*Age*), return on stock (*Ret*), return on total assets (*Roa*), leverage ratio (*Lev*), the market-to-book equity ratio (*MB*), operating income growth rate (*Growth*), firm size (*Size*), accrual earnings management (*ACC*), ownership concentration (*Top1*) and the lagged one-period negative return skewness coefficient. Detailed definitions of the variables are provided in the appendix.

3.3. Data sources

In this study, all available data of A-share listed companies and mutual funds from 2007 to 2022 are collected from the China Stock Market and Accounting Research (CSMAR) database as research samples, and the data are processed as follows: (1) *ST, ST, suspended and delisted stocks are excluded; (2) samples with less than 30 trading weeks per year are eliminated; (3) stocks from finance and insurance industries are excluded; (4) companies that have been listed less than a year are excluded; (5) remove samples with missing major variables; (6) considering extreme values, perform Winsorization on all continuous variables by trimming the top and bottom 1%. Finally, a total of 28,790 firm-year observations are obtained. Table 1 reports the descriptive statistics of the variables.

Table 1
Descriptive statistics.

Variable	N	Mean	SD	Min	P25	P50	P75	Max
FNCSKEW	28790	-0.3136	0.7063	-2.4822	-0.7036	-0.2673	0.1112	1.6777
FDUVOL	28790	-0.2099	0.4659	-1.3927	-0.5162	-0.2074	0.1028	1.0006
QFHPF	28790	0.3617	0.3276	0	0.0665	0.2829	0.5751	1
Dturn	28790	-0.3181	0.9116	-3.9784	-0.6971	-0.0692	0.3362	0.7712

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Age	28790	2.2987	0.64	1.0986	1.7918	2.3979	2.8332	3.4657
Ret	28790	-0.0014	0.0011	-0.0062	-0.0018	-0.0011	-0.0006	-0.0001
NCSKEW	28790	-0.3045	0.7022	-2.4517	-0.6892	-0.2609	0.1164	1.614
Roa	28790	0.0367	0.0569	-0.2152	0.013	0.0346	0.0636	0.194
Lev	28790	0.4471	0.1986	0.0638	0.2928	0.4466	0.5978	0.8731
MB	28790	4.0637	2.6449	1.2883	2.4297	3.3367	4.739	17.4268
Growth	28790	0.3949	1.0833	-0.689	-0.0306	0.1323	0.4188	7.8018
Size	28790	22.2616	1.2781	19.8898	21.3587	22.0835	22.9871	26.2096
ACC	28790	0.0693	0.0728	0.0008	0.0219	0.0466	0.0898	0.4018
Top1	28790	0.3404	0.1483	0.0845	0.2243	0.3183	0.4412	0.7402

4. Main empirical results

4.1. Baseline regression

This paper employs the following regression model to examine the relationship between quantitative mutual funds holdings and stock price crash risk:

$$Crash_{i,t+1} = \beta_0 + \beta_1 QFHPF_{i,t} + \gamma Control_{i,t} + \theta \sum Ind + \delta \sum Year + \varepsilon_{i,t} \quad (5)$$

where *Crash* represents the risk of stock price crash in the next year. To differentiate it from the risk of stock price crash in the current year, *FNCSKEW* or *FDUVOL* is utilized below. *Control* represents a series of control variables, and this regression also includes industry fixed effects and year fixed effects. The results of the baseline regression are shown in Table 2. Whether including partial control variables or including all control variables, the regression coefficient of *QFHPF* is significantly negative at the 1% level, which preliminarily confirms that the higher the proportion of holdings of quantitative mutual funds, the lower the stock price crash risk.

Table 2
Baseline regression.

	(1) *FNCSKEW*	(2) *FDUVOL*	(3) *FNCSKEW*	(4) *FDUVOL*
QFHPF	-0.0604***	-0.0336***	-0.0518***	-0.0311***
	(0.0144)	(0.0091)	(0.0144)	(0.0092)
Dturn	0.0078	0.0054	0.0106*	0.0081**
	(0.0058)	(0.0038)	(0.0058)	(0.0038)
Age	-0.0734***	-0.0565***	-0.0822***	-0.0562***
	(0.0076)	(0.0050)	(0.0080)	(0.0052)
Ret	-22.3239***	-9.8618***	-9.6975**	0.4714
	(4.5779)	(3.0584)	(4.7826)	(3.1937)
NCSKEW	0.0545***	0.0359***	0.0507***	0.0329***
	(0.0062)	(0.0041)	(0.0062)	(0.0041)
Roa	0.6972***	0.3523***	0.5648***	0.3035***
	(0.0762)	(0.0509)	(0.0843)	(0.0557)
Lev			-0.1914***	-0.1015***

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			(0.0333)	(0.0219)
MB			0.0198***	0.0115***
			(0.0020)	(0.0014)
Growth			0.0017	0.0014
			(0.0043)	(0.0029)
Size			0.0160***	-0.0036
			(0.0050)	(0.0033)
ACC			0.0698	0.0046
			(0.0586)	(0.0393)
Top1			-0.0980***	-0.0534**
			(0.0320)	(0.0212)
Constant	0.0354	0.0634	-0.2469**	0.1650**
	(0.0495)	(0.0386)	(0.1136)	(0.0773)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	28790	28790	28790	28790
adj. R^2	0.0519	0.0566	0.0554	0.0602

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.2. Robustness tests

4.2.1. Instrumental variable method

This subsection uses the instrumental variable method to mitigate potential endogeneity issues. We use the average value of *QFHPF* in other companies in the same industry as the instrumental variable. The results in Table 3 show that the coefficient of *QFHPF_IV* in the first stage is significantly positive at the 1% level and the *Cragg-Donald Wald F* statistic is greater than 10, which means *FHPF_IV* is not a weak instrumental variable. In the second stage, the regression coefficients of *QFHPF* are all significantly negative, that is, after addressing the potential endogeneity issues, the main conclusion of this study remain unchanged.

Table 3
Instrumental variable method.

	First stage	Second stage	
	(1)	(2)	(3)
	QFHPF	*FNCSKEW*	*FDUVOL*
QFHPF		-0.8670***	-0.6116***
		(0.3333)	(0.2215)
QFHPF_IV	0.2041***		
	(0.0316)		
Dturn	-0.0043*	0.0070	0.0056
	(0.0025)	(0.0064)	(0.0042)
Age	-0.0164***	-0.0959***	-0.0662***
	(0.0044)	(0.0104)	(0.0069)
Ret	20.4053***	6.4888	12.0551**

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	(2.3458)	(8.4732)	(5.6843)
NCSKEW	-0.0004	0.0502***	0.0327***
	(0.0029)	(0.0066)	(0.0044)
Roa	0.3713***	0.8636***	0.5167***
	(0.0418)	(0.1540)	(0.1026)
Lev	0.1308***	-0.0854	-0.0261
	(0.0185)	(0.0573)	(0.0382)
MB	-0.0092***	0.0124***	0.0062**
	(0.0010)	(0.0037)	(0.0025)
Growth	0.0002	0.0018	0.0013
	(0.0022)	(0.0047)	(0.0032)
Size	-0.0293***	-0.0074	-0.0203***
	(0.0025)	(0.0112)	(0.0074)
ACC	0.0378	0.1049*	0.0292
	(0.0268)	(0.0634)	(0.0427)
Top1	0.0552***	-0.0530	-0.0207
	(0.0172)	(0.0399)	(0.0267)
Constant	0.7389***	0.3573	0.5981***
	(0.0573)	(0.2798)	(0.1869)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Cragg-Donald Wald F statistic	53.501		
N	28724	28724	28724

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.2.2. Heckman two-stage regression

The influence of *QFHPF* on stock price crash risk may be affected by the issue of self-selection bias. To mitigate self-selection bias, Heckman two-stage regression was conducted in this subsection. In the first stage, Probit regression is conducted with a dummy variable indicating whether it is held by quantitative mutual funds (*QFHPF_D*) as the dependent variable. The independent variables include some other variables in addition to the control variables in the baseline regression, and the Inverse Mills Ratio (*IMR*) is calculated accordingly. In the second stage, *IMR* is introduced into the baseline regression as a control variable. Table 4 shows that the coefficients of *IMR* are not significant, suggesting that there is no serious self-selection bias problem. Furthermore, the coefficients of *QFHPF* are significantly negative at the 1% level, indicating that the main findings of this study remain robust after addressing the concern of self-selection bias.

Table 4

Heckman two-stage regression.

	First stage	Second stage	
	(1)	(2)	(3)
	QFHPF_D	*FNCSKEW*	*FDUVOL*
QFHPF		-0.0833***	-0.0444***
		(0.0155)	(0.0102)

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Dturn	0.0154	0.0137**	0.0089**
	(0.0149)	(0.0060)	(0.0040)
Age	-0.1555***	-0.0802***	-0.0527***
	(0.0241)	(0.0085)	(0.0056)
Ret	73.7836***	-7.4614	1.7618
	(11.9053)	(5.3962)	(3.5629)
NCSKEW	0.0792***	0.0505***	0.0334***
	(0.0153)	(0.0067)	(0.0044)
Roa	3.3360***	0.6099***	0.3012***
	(0.2142)	(0.1042)	(0.0688)
Lev	-1.5393***	-0.1585***	-0.0855***
	(0.0885)	(0.0374)	(0.0247)
MB	0.0707***	0.0209***	0.0130***
	(0.0055)	(0.0023)	(0.0015)
Growth	-0.0310***	-0.0002	0.0007
	(0.0097)	(0.0048)	(0.0032)
Size	0.6209***	0.0084	-0.0098**
	(0.0173)	(0.0072)	(0.0048)
ACC	0.0234	0.0330	-0.0360
	(0.1491)	(0.0653)	(0.0431)
Top1	-0.6223***	-0.0907***	-0.0470**
	(0.0950)	(0.0324)	(0.0214)
MER	-0.4720***		
	(0.1778)		
EqN	0.1596***		
	(0.0274)		
Inv	0.9641***		
	(0.0761)		
Wage	0.2458***		
	(0.0196)		
MH	1.1642***		
	(0.1016)		
IMR		0.0208	-0.0062
		(0.0356)	(0.0235)
Constant	-16.4792***	-0.0155	0.3262***
	(0.4049)	(0.1811)	(0.1196)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	28790	28790	28790
Pseudo R2/Wald chi2	0.3090	1587.57	1668.12

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The detailed definitions of control variables, including management expense rate (*MER*), nature of property rights (*EqN*), institutional investors' shareholding ratio (*Inv*), management compensation (*Wage*), and management shareholding ratio (*MH*), are provided in the appendix.

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4.2.3. Alternative measures of stock price crash risk

Considering the significant differences among various sub-markets (Main-Board Market, Growth Enterprises Market, and STAR Market) in the A-share market, this subsection replaces the dependent variable with *FNCSKEW* and *FDUVOL*, both calculated using the value-weighted average return of each sub-market (*FNCSKEW_d* and *FDUVOL_d*). The results in Table 5 indicate that the baseline finding remains robust.

Table 5

Alternative measures of stock price crash risk.

	(1)	(2)
	FNCSKEW_d	*FDUVOL_d*
QFHPF	-0.0500***	-0.0284***
	(0.0146)	(0.0094)
Dturn	0.0006	-0.0014
	(0.0059)	(0.0039)
Age	-0.0337***	-0.0123**
	(0.0083)	(0.0055)
Ret	-4.4501	3.8314
	(4.8982)	(3.2416)
NCSKEW	0.0391***	0.0243***
	(0.0063)	(0.0041)
Roa	0.5896***	0.3130***
	(0.0859)	(0.0572)
Lev	-0.1617***	-0.0794***
	(0.0348)	(0.0234)
MB	0.0218***	0.0134***
	(0.0021)	(0.0014)
Growth	0.0037	0.0033
	(0.0044)	(0.0029)
Size	0.0262***	0.0054
	(0.0052)	(0.0034)
ACC	0.0684	-0.0063
	(0.0599)	(0.0398)
Top1	-0.0619*	-0.0297
	(0.0340)	(0.0230)
Constant	-0.5982***	-0.1358*
	(0.1192)	(0.0775)
Industry FE	Yes	Yes
Year FE	Yes	Yes
N	28790	28790
adj. R^2	0.0450	0.0470

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The main difference between *FNCSKEW_d* and *FNCSKEW* lies in the calculation of the firm-specific

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weekly return of stock *i* in week *j* ($w_{i,j}$). When calculating *FNCSKEW_d*, if stock *i* belongs to the Growth Enterprises Market, then $r_{m,j}$ in Eq. (1) represents the value-weighted average return of the Growth Enterprises Market for week *j*. If stock *i* belongs to the STAR Market, $r_{m,j}$ in Eq. (1) represents the value-weighted average return of the STAR Market for week *j*. When calculating *FNCSKEW*, $r_{m,j}$ in Eq. (1) represents the value-weighted average return of the A-share market for week *j*. The difference between *FDUVOL_d* and *FDUVOL* is similar.

4.2.4. Exclude the impact of mutual funds

In order to verify that the inhibitory effect of quantitative mutual funds holdings (*QFHPF*) on stock price crash risk is not caused by its institutional investor nature, this subsection tests the relationship between mutual funds holdings (*FHP*) and stock price crash risk. Table 6 shows that the coefficients of *FHP* are significantly positive at the 1% level, indicating that mutual funds holdings (*FHP*) may exacerbate stock price crash risk due to herding effect. This also implies that the inhibitory effect of quantitative mutual funds holdings (*QFHPF*) on stock price crash risk is not due to its institutional investor nature.

Table 6

mutual funds holdings (*FHP*) and stock price crash risk.

	(1)	(2)
	FNCSKEW	*FDUVOL*
FHP	1.1190***	0.7086***
	(0.0661)	(0.0470)
Dturn	0.0104*	0.0080**
	(0.0058)	(0.0038)
Age	-0.0722***	-0.0499***
	(0.0079)	(0.0051)
Ret	-7.3663	1.9821
	(4.7588)	(3.1761)
NCSKEW	0.0421***	0.0275***
	(0.0062)	(0.0041)
Roa	0.2358***	0.0957*
	(0.0868)	(0.0572)
Lev	-0.1390***	-0.0681***
	(0.0330)	(0.0218)
MB	0.0127***	0.0071***
	(0.0021)	(0.0014)
Growth	0.0014	0.0012
	(0.0043)	(0.0028)
Size	-0.0037	-0.0161***
	(0.0051)	(0.0034)
ACC	0.0450	-0.0111
	(0.0587)	(0.0394)
Top1	-0.0410	-0.0171
	(0.0320)	(0.0211)

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Constant	0.0809	0.3739***
	(0.1152)	(0.0781)
Industry FE	Yes	Yes
Year FE	Yes	Yes
N	28790	28790
adj. R^2	0.0621	0.0664

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The detailed definition of *FHP* is provided in the appendix.

4.2.5. Propensity Score Matching

Compared to stocks not held by quantitative mutual funds, stocks held by quantitative mutual funds may possess certain different characteristics, which could potentially impact stock price stability. This subsection employs the propensity score matching (PSM) method to alleviate such endogeneity problems. First, the Logit model is estimated, where the dependent variable is the dummy variable *QFHPF_D*, and the independent variables include all control variables from the baseline regression. Second, the one-to-one nearest neighbor matching method is used to generate new samples, and subsequently, the baseline regression model is re-executed. The results in Table 7 show that the main findings of this paper still hold after eliminating the systematic differences between the treatment group and the control group.

Table 7

Propensity Score Matching.

	(3)	(4)
	FNCSKEW	*FDUVOL*
QFHPF	-0.0511***	-0.0312***
	(0.0145)	(0.0092)
Dturn	0.0095	0.0074*
	(0.0060)	(0.0039)
Age	-0.0819***	-0.0561***
	(0.0082)	(0.0053)
Ret	-10.0690**	0.1545
	(4.8441)	(3.2239)
NCSKEW	0.0502***	0.0324***
	(0.0063)	(0.0041)
Roa	0.5544***	0.2970***
	(0.0857)	(0.0565)
Lev	-0.1868***	-0.0939***
	(0.0344)	(0.0225)
MB	0.0194***	0.0108***
	(0.0021)	(0.0014)
Growth	0.0012	0.0009
	(0.0043)	(0.0029)
Size	0.0152***	-0.0046
	(0.0056)	(0.0036)

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ACC	0.0779	0.0120
	(0.0599)	(0.0400)
Top1	-0.1039***	-0.0552**
	(0.0327)	(0.0214)
Constant	-0.2243*	0.1885**
	(0.1235)	(0.0820)
Industry FE	Yes	Yes
Year FE	Yes	Yes
N	28072	28072
adj. R^2	0.0550	0.0597

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3. Cross-sectional heterogeneity

4.3.1. Management characteristics

Some special experiences of the senior management of listed companies may affect the stock price crash risk (Cui et al., 2022; Wang et al., 2022), so it is interesting to examine whether the impact of quantitative mutual funds holdings on the stock price crash risk is different among listed companies with different senior management backgrounds. In this subsection, according to whether the senior management of listed companies has overseas background, the sample data is divided into two groups (MOB and Non-MOB) for regression, and the results are shown in Table 8. Thanks to the powerful information search and information processing capabilities of quantitative investment technology, it can effectively identify some factors conducive to stock price stability in listed companies and better exert their effectiveness, making the inhibitory effect of quantitative mutual funds holdings on stock price crash risk more significant in the MOB subsample.

Table 8

Management overseas background.

	FNCSKEW		*FDUVOL*	
	(1)	(2)	(3)	(4)
	Non-MOB	MOB	Non-MOB	MOB
QFHPF	-0.0203	-0.0826***	-0.0124	-0.0489***
	(0.0207)	(0.0198)	(0.0132)	(0.0127)
Dturn	0.0128	0.0090	0.0082	0.0081
	(0.0090)	(0.0078)	(0.0058)	(0.0051)
Age	-0.0829***	-0.0822***	-0.0556***	-0.0569***
	(0.0120)	(0.0109)	(0.0078)	(0.0071)
Ret	-5.9905	-11.1985*	4.5702	-2.1177
	(7.2017)	(6.3729)	(4.7420)	(4.3444)
NCSKEW	0.0556***	0.0413***	0.0329***	0.0299***
	(0.0092)	(0.0085)	(0.0060)	(0.0056)
Roa	0.5752***	0.5429***	0.2791***	0.3126***
	(0.1286)	(0.1109)	(0.0844)	(0.0735)
Lev	-0.1890***	-0.1977***	-0.1067***	-0.1001***

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	(0.0500)	(0.0438)	(0.0333)	(0.0290)
MB	0.0168***	0.0233***	0.0101***	0.0134***
	(0.0031)	(0.0027)	(0.0021)	(0.0018)
Growth	-0.0001	0.0039	-0.0009	0.0034
	(0.0059)	(0.0065)	(0.0040)	(0.0042)
Size	0.0124	0.0178***	-0.0062	-0.0027
	(0.0079)	(0.0065)	(0.0053)	(0.0043)
ACC	-0.0093	0.1130	-0.0539	0.0411
	(0.0854)	(0.0809)	(0.0571)	(0.0544)
Top1	-0.0737	-0.1181***	-0.0483	-0.0542*
	(0.0471)	(0.0423)	(0.0307)	(0.0285)
Constant	-0.1516	-0.3303**	0.2191*	0.1250
	(0.1763)	(0.1502)	(0.1187)	(0.1027)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	13067	15723	13067	15723
adj. R^2	0.0621	0.0522	0.0677	0.0563
Subsample comparison of	Chi-Square = 4.83		Chi-Square = 4.09	
coefficients on QFHPF	P-value = 0.0280		P-value = 0.0431	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3.2. Internal governance efficiency

Generally speaking, companies with higher internal governance efficiency can better restrain the self-interest behavior of management, allowing their disclosed information to objectively reflect the true operating condition of the company. High-quality information sources provide conditions for quantitative investment techniques to fully exert their effectiveness. Therefore, we predict that the negative relationship between quantitative mutual funds holdings and stock price crash risk is more significant in the subsample with higher internal governance efficiency. In Table 9, this subsection divides the enterprises whose CEO does not also serve as Chairman into the subsample with high internal governance efficiency and the enterprises whose CEO also serves as Chairman into the subsample with poor internal governance efficiency. The regression results are consistent with our prediction.

Table 9
CEO-Chairman Duality.

	FNCSKEW		*FDUVOL*	
	(1)	(2)	(3)	(4)
	NOT-DUAL	DUAL	NOT-DUAL	DUAL
QFHPF	-0.0708***	0.0037	-0.0433***	0.0055
	(0.0166)	(0.0297)	(0.0106)	(0.0186)
Dturn	0.0116*	0.0058	0.0066	0.0111
	(0.0067)	(0.0122)	(0.0044)	(0.0078)
Age	-0.0813***	-0.0727***	-0.0572***	-0.0522***
	(0.0093)	(0.0173)	(0.0061)	(0.0111)

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Ret	-8.4243	-12.3210	0.0112	1.3813
	(5.5811)	(9.5886)	(3.6908)	(6.3575)
NCSKEW	0.0517***	0.0409***	0.0321***	0.0319***
	(0.0070)	(0.0135)	(0.0047)	(0.0087)
Roa	0.5035***	0.7302***	0.2914***	0.3554***
	(0.0988)	(0.1696)	(0.0657)	(0.1102)
Lev	-0.1865***	-0.1999***	-0.1010***	-0.0982**
	(0.0385)	(0.0679)	(0.0254)	(0.0442)
MB	0.0191***	0.0221***	0.0115***	0.0120***
	(0.0024)	(0.0040)	(0.0016)	(0.0026)
Growth	0.0014	0.0037	0.0012	0.0018
	(0.0047)	(0.0100)	(0.0033)	(0.0063)
Size	0.0140**	0.0248**	-0.0040	-0.0007
	(0.0056)	(0.0111)	(0.0038)	(0.0068)
ACC	0.0417	0.1646	-0.0165	0.0648
	(0.0671)	(0.1215)	(0.0454)	(0.0809)
Top1	-0.0901**	-0.1149*	-0.0419*	-0.0787*
	(0.0367)	(0.0660)	(0.0243)	(0.0436)
Constant	-0.2240*	-0.2183	0.1610*	0.2253
	(0.1259)	(0.2657)	(0.0870)	(0.1570)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	21794	6996	21794	6996
adj. R^2	0.0580	0.0414	0.0635	0.0453
Subsample comparison of coefficients on QFHPF	Chi-Square = 5.15 P-value = 0.0232		Chi-Square = 5.53 P-value = 0.0187	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3.3. Information environment

A favorable external information environment helps investors quickly extract valuable information from the noisy capital market, thereby alleviating information asymmetry and reducing the risk of stock price crash. Financial analysts' professional interpretation of the public information of listed companies is conducive to high-quality information dissemination, thus optimizing the market information environment. Therefore, more analyst coverage also means a better information environment. We divide the sample into two sub-samples according to the level of analyst coverage. Table 10 shows that the inhibitory effect of quantitative mutual funds holdings on stock price crash risk is more significant in the subsample with high analyst coverage. This suggests that a good information environment helps quantitative investment to conduct efficient data mining, which in turn promotes stock price stability.

Table 10

Analyst Coverage.

	FNCSKEW		*FDUVOL*	
	(1)	(2)	(3)	(4)

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	Low	High	Low	High
QFHPF	-0.0298*	-0.1159***	-0.0218**	-0.0621***
	(0.0173)	(0.0261)	(0.0107)	(0.0175)
Dturn	0.0052	0.0072	0.0083	0.0018
	(0.0083)	(0.0081)	(0.0053)	(0.0055)
Age	-0.0663***	-0.0625***	-0.0490***	-0.0411***
	(0.0116)	(0.0114)	(0.0073)	(0.0079)
Ret	-12.2330*	-9.5056	-1.1676	1.2689
	(6.3824)	(7.1193)	(4.1280)	(4.9935)
NCSKEW	0.0502***	0.0334***	0.0316***	0.0229***
	(0.0082)	(0.0093)	(0.0053)	(0.0063)
Roa	0.3141***	0.3316**	0.1891***	0.1032
	(0.1136)	(0.1451)	(0.0731)	(0.0975)
Lev	-0.0765	-0.2301***	-0.0095	-0.1613***
	(0.0467)	(0.0495)	(0.0297)	(0.0346)
MB	0.0097***	0.0250***	0.0036*	0.0182***
	(0.0031)	(0.0030)	(0.0020)	(0.0021)
Growth	0.0068	-0.0055	0.0047	-0.0040
	(0.0054)	(0.0073)	(0.0034)	(0.0052)
Size	-0.0224**	0.0047	-0.0289***	-0.0072
	(0.0089)	(0.0072)	(0.0056)	(0.0049)
ACC	0.0346	0.0502	0.0058	-0.0338
	(0.0801)	(0.0859)	(0.0524)	(0.0606)
Top1	-0.0865*	-0.0874**	-0.0483*	-0.0473
	(0.0454)	(0.0442)	(0.0289)	(0.0309)
Constant	0.4888**	0.0349	0.6467***	0.2717**
	(0.1988)	(0.1651)	(0.1224)	(0.1207)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	15661	13129	15661	13129
adj. R^2	0.0471	0.0695	0.0558	0.0751
Subsample comparison of coefficients on QFHPF	Chi-Square = 7.75 P-value = 0.0054		Chi-Square = 4.01 P-value = 0.0452	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Analyst Coverage is measured by the natural logarithm of the number of analysts (teams) that have followed the listed company each year. This table is grouped based on the annual industry median of Analyst Coverage, with samples exceeding the annual industry median categorized as the High Analyst Coverage group, and the remaining samples categorized as the Low Analyst Coverage group.

4.3.4. Profit Volatility

Generally speaking, companies with stable operating profits tend to have relatively stable stock prices, while companies with larger fluctuations in operating profits are more likely to magnify investors' greed or panic, prompting them to make irrational investment decisions and leading to sharp increases or declines in stock prices. Quantitative investment is more rational than

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discretionary investment, so it is reasonable to believe that the negative effect of quantitative mutual funds holdings on stock price crash risk is stronger in companies with high profit volatility. In view of this, this subsection divides the sample into two subsamples for verification according to the profit volatility, and the results in Table 11 are consistent with the above reasoning.

Table 11

Profit Volatility.

	FNCSKEW		*FDUVOL*	
	(1)	(2)	(3)	(4)
	Low	High	Low	High
QFHPF	-0.0188	-0.0826***	-0.0107	-0.0495***
	(0.0203)	(0.0203)	(0.0130)	(0.0129)
Dturn	0.0128	0.0082	0.0116**	0.0043
	(0.0079)	(0.0085)	(0.0052)	(0.0056)
Age	-0.0788***	-0.0827***	-0.0552***	-0.0558***
	(0.0111)	(0.0112)	(0.0072)	(0.0073)
Ret	-7.1444	-11.7798*	2.1113	-1.9014
	(7.3038)	(6.4233)	(4.7952)	(4.3069)
NCSKEW	0.0523***	0.0466***	0.0349***	0.0301***
	(0.0086)	(0.0090)	(0.0057)	(0.0058)
Roa	0.9245***	0.4997***	0.4541***	0.2584***
	(0.1989)	(0.0969)	(0.1320)	(0.0645)
Lev	-0.2011***	-0.1619***	-0.1200***	-0.0834***
	(0.0493)	(0.0460)	(0.0334)	(0.0300)
MB	0.0200***	0.0187***	0.0114***	0.0114***
	(0.0033)	(0.0026)	(0.0022)	(0.0017)
Growth	-0.0058	0.0070	-0.0041	0.0051
	(0.0069)	(0.0055)	(0.0044)	(0.0037)
Size	0.0147**	0.0168**	-0.0035	-0.0033
	(0.0070)	(0.0069)	(0.0046)	(0.0046)
ACC	0.0952	0.0333	0.0110	0.0029
	(0.0953)	(0.0749)	(0.0643)	(0.0502)
Top1	-0.0703	-0.1313***	-0.0395	-0.0703**
	(0.0441)	(0.0440)	(0.0293)	(0.0290)
Constant	-0.2919*	-0.1975	0.1712	0.1456
	(0.1593)	(0.1598)	(0.1060)	(0.1108)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	14657	14133	14657	14133
adj. R^2	0.0548	0.0564	0.0617	0.0591
Subsample comparison of coefficients on QFHPF	Chi-Square = 5.06 P-value = 0.0245		Chi-Square = 4.62 P-value = 0.0316	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Profit Volatility is measured by the standard deviation of the ratio of *EBIT* (Earnings Before Interest and Taxes)

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to total assets of the listed company over the past three years. This table is grouped based on the annual industry median of Profit Volatility, with samples exceeding the annual industry median categorized as the High Profit Volatility group, and the remaining samples categorized as the Low Profit Volatility group.

4.4. Further analysis

Studying the mechanism by which quantitative mutual funds holdings affect the risk of stock price crash is of great importance. On the one hand, quantitative investment technology possesses extraordinary capabilities in information collection, conversion, and processing. This enables stock prices to more promptly reflect specific market information, thereby allowing quantitative mutual funds holdings to mitigate stock price crash risk by reducing volatility. On the other hand, quantitative investment technology can achieve more accurate assessment of stock prices by introducing cutting-edge algorithms. Meanwhile, its more rational investment style compared with discretionary investment can also help avoid speculative bubbles caused by excessive greed. Therefore, we infer that quantitative mutual funds holdings can restrain stock price crash risk by stabilizing stock valuation. In this subsection, stock return volatility (*Sigma*) and stock valuation (*TobinQ*) are employed as mediating variables and the following three regressions are used to examine the possible mechanisms.

$$Crash_{i,t+1} = \beta_0 + \beta_1 QFHPF_{i,t} + \gamma Control_{i,t} + \theta \sum Ind + \delta \sum Year + \varepsilon_{i,t} \quad (6)$$

$$Mediator_{i,t} = \alpha_0 + \alpha_1 QFHPF_{i,t} + \gamma Control_{i,t} + \theta \sum Ind + \delta \sum Year + \tau_{i,t} \quad (7)$$

$$Crash_{i,t+1} = \beta'_0 + \beta'_1 Mediator_{i,t} + \beta'_2 QFHPF_{i,t} + \gamma Control_{i,t} + \theta \sum Ind + \delta \sum Year + \omega_{i,t} \quad (8)$$

Table 12 shows that higher *QFHPF* is associated with lower *Sigma*, which means lower risk of stock price crash, and the proportion of total effect that is mediated by *Sigma* is 6.4% and 5.2%. Similarly, Table 13 shows that higher *QFHPF* is associated with lower *TobinQ*, which indicates lower stock price crash risk, and the proportion of total effect that is mediated by *TobinQ* is 3.7% and 2.7%.

Table 12

Possible mechanism: *Sigma*.

	(1) *FNCSKEW*	(2) *Sigma*	(3) *FNCSKEW*	(4) *FDUVOL*	(5) *FDUVOL*
QFHPF	-0.0518***	-0.0005***	-0.0485***	-0.0311***	-0.0295***
	(0.0144)	(0.0001)	(0.0144)	(0.0092)	(0.0092)
Sigma			6.3087***		3.0857***
			(1.0655)		(0.6930)
Dturn	0.0106*	0.0014***	0.0018	0.0081**	0.0038
	(0.0058)	(0.0000)	(0.0060)	(0.0038)	(0.0039)
Age	-0.0822***	-0.0008***	-0.0769***	-0.0562***	-0.0536***
	(0.0080)	(0.0001)	(0.0081)	(0.0052)	(0.0053)
Ret	-9.6975**	-15.4346***	87.6740***	0.4714	48.0984***

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	(4.7826)	(0.0573)	(16.8445)	(3.1937)	(11.0532)
NCSKEW	0.0507***	0.0003***	0.0490***	0.0329***	0.0321***
	(0.0062)	(0.0000)	(0.0062)	(0.0041)	(0.0041)
Roa	0.5648***	0.0021***	0.5517***	0.3035***	0.2971***
	(0.0843)	(0.0005)	(0.0842)	(0.0557)	(0.0558)
Lev	-0.1914***	0.0019***	-0.2032***	-0.1015***	-0.1073***
	(0.0333)	(0.0002)	(0.0333)	(0.0219)	(0.0220)
MB	0.0198***	0.0000***	0.0195***	0.0115***	0.0114***
	(0.0020)	(0.0000)	(0.0020)	(0.0014)	(0.0014)
Growth	0.0017	0.0000	0.0017	0.0014	0.0014
	(0.0043)	(0.0000)	(0.0043)	(0.0029)	(0.0029)
Size	0.0160***	-0.0007***	0.0205***	-0.0036	-0.0014
	(0.0050)	(0.0000)	(0.0050)	(0.0033)	(0.0033)
ACC	0.0698	0.0022***	0.0559	0.0046	-0.0022
	(0.0586)	(0.0003)	(0.0586)	(0.0393)	(0.0393)
Top1	-0.0980***	-0.0003	-0.0963***	-0.0534**	-0.0525**
	(0.0320)	(0.0002)	(0.0320)	(0.0212)	(0.0211)
Constant	-0.2469**	0.0467***	-0.5417***	0.1650**	0.0208
	(0.1136)	(0.0008)	(0.1219)	(0.0773)	(0.0830)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
N	28790	28790	28790	28790	28790
adj. R^2	0.0554	0.9568	0.0566	0.0602	0.0609
Sobel Test		-4.580***		-3.799***	
Percentage		6.4%		5.2%	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The detailed definition of *Sigma* is provided in the appendix.

Table 13

Possible mechanism: *TobinQ*.

	(1)	(2)	(3)	(4)	(5)
	FNCSKEW	*TobinQ*	*FNCSKEW*	*FDUVOL*	*FDUVOL*
QFHPF	-0.0518***	-0.0676***	-0.0499***	-0.0311***	-0.0303***
	(0.0144)	(0.0135)	(0.0144)	(0.0092)	(0.0092)
TobinQ			0.0280***		0.0125**
			(0.0077)		(0.0051)
Dturn	0.0106*	-0.0267***	0.0114*	0.0081**	0.0084**
	(0.0058)	(0.0042)	(0.0058)	(0.0038)	(0.0038)
Age	-0.0822***	0.0560***	-0.0838***	-0.0562***	-0.0569***
	(0.0080)	(0.0100)	(0.0080)	(0.0052)	(0.0052)
Ret	-9.6975**	-66.2930***	-7.8382	0.4714	1.3019
	(4.7826)	(6.3502)	(4.8266)	(3.1937)	(3.2218)
NCSKEW	0.0507***	0.0158***	0.0502***	0.0329***	0.0327***
	(0.0062)	(0.0053)	(0.0062)	(0.0041)	(0.0041)

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Roa	0.5648***	2.2370***	0.5020***	0.3035***	0.2754***
	(0.0843)	(0.1327)	(0.0870)	(0.0557)	(0.0572)
Lev	-0.1914***	-3.6473***	-0.0891**	-0.1015***	-0.0558*
	(0.0333)	(0.0622)	(0.0437)	(0.0219)	(0.0286)
MB	0.0198***	0.4058***	0.0084**	0.0115***	0.0065***
	(0.0020)	(0.0085)	(0.0038)	(0.0014)	(0.0025)
Growth	0.0017	-0.0055	0.0019	0.0014	0.0015
	(0.0043)	(0.0051)	(0.0043)	(0.0029)	(0.0029)
Size	0.0160***	-0.0431***	0.0172***	-0.0036	-0.0031
	(0.0050)	(0.0083)	(0.0050)	(0.0033)	(0.0033)
ACC	0.0698	0.0181	0.0693	0.0046	0.0043
	(0.0586)	(0.0662)	(0.0586)	(0.0393)	(0.0393)
Top1	-0.0980***	-0.1146***	-0.0948***	-0.0534**	-0.0519**
	(0.0320)	(0.0422)	(0.0322)	(0.0212)	(0.0212)
Constant	-0.2469**	2.8502***	-0.3269***	0.1650**	0.1293
	(0.1136)	(0.1961)	(0.1157)	(0.0773)	(0.0786)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
N	28790	28790	28790	28790	28790
adj. R^2	0.0554	0.8147	0.0559	0.0602	0.0604
Sobel Test		-3.283***		-2.413**	
Percentage		3.7%		2.7%	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The detailed definition of *TobinQ* is provided in the appendix.

5. Conclusion

Different from the limited computing power and attention of the human brain, the computing power of quantitative investment strategies is approaching infinity with the continuous improvement of computer hardware and the rapid development of digital technology. This enables it to collect and process noisy capital market information more efficiently. Furthermore, quantitative investment, as specific computer programs developed by incorporating previous investment experience, possess the capability to make optimal trading decisions that align with the risk-return targets of the strategies in complex information environments, without being swayed by investors' subjective preferences. This renders its investment more rational and effectively prevents speculative bubbles caused by excessive greed. From the perspective of quantitative mutual funds holdings, this study finds that the widespread application of quantitative investment technology can suppress stock price crash risk by smoothing the volatility of stock returns and optimizing the valuation of firms. Moreover, this effect is more pronounced in subsamples characterized by executives with overseas backgrounds, higher internal governance efficiency, greater analyst attention, and higher profit volatility. Our findings confirm the positive role of quantitative investment in stabilizing stock prices, providing evidence supporting the widespread application of financial technology and cutting-edge algorithms in the investment community.

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CRediT authorship contribution statement

Hongyin Li: Conceptualization, Data curation, Methodology, Software, Validation, Writing - original draft, Writing - review & editing. **Tiancheng Li:** Supervision, Validation, Writing - review & editing. **Zhichao Zhang:** Investigation, Supervision.

Declaration of Competing Interest

None

Appendix. Definitions of variables

NCSKEW	The negative skewness of firm-specific weekly returns over the fiscal year.
DUVOL	The log of the ratio of the standard deviations of down-week to up-week firm-specific weekly returns.
QFHPF	The ratio of shares held by quantitative mutual funds to the total shares held by all mutual funds.
Dturn	The difference between the stock turnover rate in year t and the stock turnover rate in year t-1, divided by the stock turnover rate in year t.
Age	The natural logarithm of the number of years a firm has been listed plus one.
Ret	Mean of the stock's firm-specific weekly return.
Roa	Returns on total assets.
Lev	Total debts divided by total assets.
MB	The market value of equity divided by the book value of equity.
Growth	The difference between the current year's operating income and the prior year's operating income, divided by the prior year's operating income.
Size	The natural logarithm of market capitalization.
ACC	The absolute value of discretionary accruals, where discretionary accruals are estimated from the modified Jones model.
Top1	Shareholding ratio of the largest shareholder.
QFHPF_D	A dummy variable takes a value of 1 when listed company is held by quantitative mutual funds, and 0 otherwise.
MER	Management expenses over total assets.
EqN	A dummy variable, where the value is 1 for state-owned enterprises and 0 for non-state-owned enterprises.
Inv	The proportion of shares of listed companies held by institutional investors.
Wage	The total compensation of top three senior executives in listed companies.
MH	The ratio of the number of shares held by senior management to the total number of shares in the listed company.
FHP	The ratio of the number of shares held by mutual funds to the total number of shares of listed companies.
Sigma	Standard deviation of the stock's firm-specific weekly return.
TobinQ	Total market value divided by total assets.

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Quantitative investment and stock price crash risk in China: Perspective of quantitative mutual funds holdings

ABSTRACT

This study examines the impact of quantitative investment on stock price crash risk from the perspective of quantitative mutual funds holdings. The results show that quantitative mutual funds holdings can significantly reduce stock price crash risk, and this effect is more pronounced in subsamples characterized by executives with overseas backgrounds, higher internal governance efficiency, greater analyst attention, and higher profit volatility. Further research finds that quantitative mutual funds holdings can suppress the risk of stock price crash by smoothing the volatility of stock returns and optimizing the valuation of firms. This study sheds light on the effects of quantitative investment on stock price crash risk.

Keywords: Quantitative investment; Quantitative mutual funds; Stock price crash risk; China

1. Introduction

Quantitative investment refers to a trading strategy that employs quantitative methodologies and computer algorithms to develop trading models and execute buying and selling orders, aiming to achieve consistent and stable returns. In recent years, the capital market has witnessed a rapid accumulation of various types of data due to the rapid development of digital technology. To cope with the complex, high-noise, and non-linear nature of market data, artificial intelligence algorithms like machine learning have been implemented which has strongly driven the advancement of quantitative investment, especially in the domain of quantitative mutual funds. Quantitative investment is deemed more rational and efficient when compared to discretionary investment. Consequently, the widespread implementation of quantitative investment technology is expected to influence investors' investment philosophy, subsequently impacting stock price dynamics.

Despite the substantial literature on quantitative investment research, the primary focus of these studies lies in building more effective quantitative trading strategies (Fang et al., 2022) or determining whether the introduction of new algorithms can improve the profitability and prediction accuracy of such strategies (Wang et al., 2022). There is little literature examining the impact of the widespread application of quantitative investment techniques on stock price stability. Quantitative investment, as a highly specialized investment strategy, demands not only proficiency in financial, statistical, and programming knowledge from investors, but also relies heavily on extensive data resources and sophisticated computer hardware. The combination of a high entry barrier and intricate technical demands poses significant challenges for the majority of retail investors in engaging with this field. However, with a low entry threshold for investment amounts, quantitative mutual funds, as a professional investment institution highly regarded by investors, offer retail investors an excellent opportunity to participate in quantitative investment practices. That is why this paper wants to investigate the impact of quantitative investment on stock price crash risk from the perspective of quantitative mutual funds holdings.

According to the theory of behavioral finance, some cognitive biases, such as herding bias (Li and Jiang, 2022; Deng et al., 2018) and over-confidence bias (Kim et al., 2016), contribute

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significantly to stock price crash. Discretionary investors are susceptible to the influence of instantaneous emotions, which often leads to irrational decision-making. On the contrary, quantitative trading strategies, as specific computer programs developed by learning historical data and summarizing trading experience, can firmly implement the established trading intention and make scientific trading decisions to overcome the interference of cognitive bias on investment. Therefore, the extensive application of quantitative investment technology can effectively contribute to stock price stabilization.

The empirical findings show that an increase in the shareholding ratio of quantitative mutual funds is associated with a decrease in the risk of stock price crash. Moreover, this inhibitory effect is more pronounced in subsamples characterized by executives with overseas backgrounds, higher internal governance efficiency, greater analyst attention, and higher profit volatility. Further research finds that quantitative mutual funds holdings can suppress the risk of stock price crash by smoothing the volatility of stock returns and optimizing the valuation of enterprises.

The main contribution of this study is threefold. First, this paper expands the research of quantitative investment. The existing literature primarily focuses on investigating new algorithms and models' effectiveness in predicting stock returns (Hu et al., 2023), as well as how to apply documented indicators to build more profitable quantitative trading strategies (Fang et al., 2022), disregarding the influence of quantitative investment on the capital market's dynamics. Second, this is the first time for the academic community to investigate the impact of quantitative mutual funds on the capital market. Third, our study enriches the relevant literature on stock price crash risk. It reveals the significance of quantitative investment in fostering the growth of the capital market and ensuring the stability of stock prices, thereby providing valuable evidence supporting the widespread application of quantitative investment and financial technology in the investment industry.

2. Prior literature and hypotheses development

Some scholars attempt to explain the risk of stock price crash based on the theory of bad news hoarding. Given the market's limited tolerance for negative news, managers tend to hide or delay the disclosure of negative information for the motivation of maintaining career prospects and seeking promotion (Graham et al., 2005; Piotroski et al., 2015; Chang et al., 2017). When negative news erupts in great quantities, the stock price experiences a collapse. For example, Kong et al. (2023) found that the uncertain news related to COVID-19 exacerbated the risk of stock price crash. Some scholars also try to use behavioral finance theory to attribute stock price crash. For instance, the study conducted by Kim et al. (2016) revealed a positive correlation between overconfident CEOs and the risk of stock price crash. Deng et al. (2018) found that mutual funds herding amplifies stock price crash risk. Jang and Kang (2019) suggest that overpricing driven by institutional investment will lead to stock price crash.

Quantitative investment has been explored in academia and industry for many years, from early multi-factor models to machine learning algorithms that are now widely ongoing discussion. Due to the high dimensionality and low signal-to-noise ratio of stock data, traditional quantitative trading models are becoming increasingly difficult to generate excess returns. As a result, the investment community is eager for the profits that come with cutting-edge technological innovation (Fischer and Krauss, 2018). Simultaneously, certain research findings also provide empirical support for the

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application of the new algorithm in quantitative investment. For example, Gu et al. (2020) showing that machine learning methods perform observably better than traditional linear regression models. Hu et al. (2023) found that machine learning yields favorable investment returns in simulated quantitative trading of China's A-share market. Therefore, an increasing number of cutting-edge techniques are being employed in quantitative investment.

Quantitative investment possesses two primary advantages in comparison to discretionary investment. Firstly, quantitative investment can collect information more extensively and process data more efficiently. Different from the limited computing power and attention of the human brain, the computing power of quantitative investment strategies is approaching infinity with the continuous improvement of computer hardware and the rapid development of digital technology. This enables it to apply big data technology to collect multi-level information from a broader dimension through the internet. Furthermore, by utilizing advanced information processing technology, quantitative investment has the capability to convert unstructured data such as images, audios, and videos into structured data and conduct real-time monitoring of market information, thereby enhancing investment efficiency and accuracy. Typically, information lag or asymmetry serves as a significant factor contributing to fluctuations in stock prices. The decline in investors' ability to access information online will increase the risk of stock price crash (Xu et al., 2021). By leveraging its powerful capabilities in information collection and processing, quantitative investment can facilitate the efficient and timely integration of heterogeneous information into stock prices, thereby reducing stock price volatility. Based on this analysis, we argue that quantitative investment can reduce the risk of stock price crash by smoothing stock price fluctuation.

Secondly, quantitative investment can overcome cognitive biases and irrational behaviors in discretionary investment. Given the limited rationality of human beings and their susceptibility to various cognitive biases (Linnainmaa et al., 2021; Bazley et al., 2021), it is susceptible for them to make irrational investment decisions when influenced by specific situations or information flows. For example, Lwin et al. (2020) found that investor sentiment in the stock market can be easily affected by different news of COVID-19. Especially when investors are driven by over-optimistic sentiment, they often tend to propel stock prices to surge significantly in a short period of time, resulting in stock prices being severely overestimated. However, once the adverse information concealed by the management is revealed, investors' sentiment is likely to quickly shift from over-optimism to over-pessimism, triggering the burst of the stock price bubble and ultimately leading to a crash. By contrast, quantitative investment strategies, which are specific computer programs developed by incorporating previous investment experience, have the ability to make optimal trading decisions that align with the risk-return targets of the strategies in complex information environments, without being influenced by investors' subjective preferences. For instance, Chen and Ren (2022) provide evidence that AI-powered funds can overcome some of the prevalent cognitive biases such as disposition effect and rank effect. Moreover, quantitative investment strategies have the potential to continue growing and evolving, especially with the introduction of cutting-edge algorithms such as machine learning. Quantitative investment, leveraging its objective and rational investment style coupled with the precise analysis of cutting-edge algorithms, can provide a more accurate valuation for stocks, thus effectively preventing the emergence of speculative bubbles. Based on this analysis, it is reasonable to believe that quantitative investment can mitigate the risk of stock price crash by optimizing the valuation of firms.

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3. Data and variables

3.1. Independent variable

The key independent variable quantitative mutual funds holdings (*QFHPF*) is constructed through the following steps. First, select the mutual funds that contain one or more of the following keywords in their investment strategy text descriptions: “Quantitative Investment”, “Quantitative Indicators”, “Quantitative Tools”, “Quantitative Model”, “Quantitative Hedge”, “Quantitative Strategy”, “Quantitative Team”, “Quantitative Multifactor”, “Quantitative Timing”, “Quantitative Stock Selection”, “Quantitative Trading”, “Quantitative Method”, “Quantitative Analysis”, “Quantitative Processing”, “Quantitative Screening”, “Quantitative Risk”, “Quantitative Modeling”, “Quantitative Multi-Strategy”, “Quantitative Pricing”, “Quantitative Valuation”, “Algorithm”, “Statistical Arbitrage”, “Backtest”, “Market Neutrality”, “Event-Driven”, “Artificial Intelligence”, “Machine Learning”, “Neural Network”, “Deep Learning”, “Big Data”, “Natural Language Processing”. Second, manually exclude mutual funds that invest in the artificial intelligence technology industry chain rather than utilizing artificial intelligence technology for investment, and categorize these screened mutual funds as quantitative mutual funds. Third, according to the listed company's funds holdings data, calculate the total number of shares held by all mutual funds and quantitative mutual funds at the end of each year during the sample period. Given the potential impact of institutional investors' behavior on the likelihood of stock price crash (Deng et al., 2018; Jang and Kang, 2019), the proxy variable for quantitative mutual funds holdings in this paper is defined as the ratio of shares held by quantitative mutual funds to the total shares held by all mutual funds.

3.2. Dependent variable and control variables

Following the literature (Hutton et al., 2009; Kim et al., 2011; Zuo et al., 2022), this study uses the negative return skewness coefficient (*NCSKEW*) and down-to-up volatility of returns (*DUVOL*) to measure stock price crash risk. The calculation process of *NCSKEW* is shown in Eq. (1), (2) and (3). Where $r_{i,j}$ is the return of stock *i* in week *j*, and $r_{m,j}$ is the value-weighted average return of A-share market in week *j*. Considering asynchronous trading, $r_{m,j-2}$ and $r_{m,j-1}$ denote the lagged terms, $r_{m,j+1}$ and $r_{m,j+2}$ denote the leading terms. Additionally, $w_{i,j}$ denotes the firm-specific weekly return of stock *i* in week *j*, and $\varepsilon_{i,j}$ is the residual term of Eq. (1). In Eq. (3), *n* represents the number of effective trading weeks for stock *i* in year *t*. The calculation process of *DUVOL* is shown in Eq. (1), (2) and (4). In addition to the variables mentioned above, the calculation of *DUVOL* requires grouping based on whether the firm-specific weekly return is above the annual mean. Weeks with returns above the annual mean are designated as up weeks, while weeks with returns below the annual mean are designated as down weeks. In Eq. (4), n_u and n_d represent the number of up weeks and down weeks for stock *i* in year *t*, respectively. Larger values of *NCSKEW* and *DUVOL* imply higher stock price crash risk.

$$r_{i,j} = \alpha + \beta_1 r_{m,j-2} + \beta_2 r_{m,j-1} + \beta_3 r_{m,j} + \beta_4 r_{m,j+1} + \beta_5 r_{m,j+2} + \varepsilon_{i,j} \quad (1)$$

$$w_{i,j} = \ln(1 + \varepsilon_{i,j}) \quad (2)$$

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$$NCSKEW_{i,t} = -\frac{n(n-1)^{\frac{3}{2}}\sum w_{i,j}^3}{(n-1)(n-2)\left(\sum w_{i,j}^2\right)^{\frac{3}{2}}} \quad (3)$$

$$DUVOL_{i,t} = ln\left[\frac{n_u - 1}{n_d - 1}\frac{\sum_{Down} w_{i,j}^2}{\sum_{Up} w_{i,j}^2}\right] \quad (4)$$

Control variables include: change in turnover rate (*Dturn*), firm age (*Age*), return on stock (*Ret*), return on total assets (*Roa*), leverage ratio (*Lev*), the market-to-book equity ratio (*MB*), operating income growth rate (*Growth*), firm size (*Size*), accrual earnings management (*ACC*), ownership concentration (*Top1*) and the lagged one-period negative return skewness coefficient. Detailed definitions of the variables are provided in the appendix.

3.3. Data sources

In this study, all available data of A-share listed companies and mutual funds from 2007 to 2022 are collected from the China Stock Market and Accounting Research (CSMAR) database as research samples, and the data are processed as follows: (1) *ST, ST, suspended and delisted stocks are excluded; (2) samples with less than 30 trading weeks per year are eliminated; (3) stocks from finance and insurance industries are excluded; (4) companies that have been listed less than a year are excluded; (5) remove samples with missing major variables; (6) considering extreme values, perform Winsorization on all continuous variables by trimming the top and bottom 1%. Finally, a total of 28,790 firm-year observations are obtained. Table 1 reports the descriptive statistics of the variables.

Table 1
Descriptive statistics.

Variable	N	Mean	SD	Min	P25	P50	P75	Max
FNCSKEW	28790	-0.3136	0.7063	-2.4822	-0.7036	-0.2673	0.1112	1.6777
FDUVOL	28790	-0.2099	0.4659	-1.3927	-0.5162	-0.2074	0.1028	1.0006
QFHPF	28790	0.3617	0.3276	0	0.0665	0.2829	0.5751	1
Dturn	28790	-0.3181	0.9116	-3.9784	-0.6971	-0.0692	0.3362	0.7712
Age	28790	2.2987	0.64	1.0986	1.7918	2.3979	2.8332	3.4657
Ret	28790	-0.0014	0.0011	-0.0062	-0.0018	-0.0011	-0.0006	-0.0001
NCSKEW	28790	-0.3045	0.7022	-2.4517	-0.6892	-0.2609	0.1164	1.614
Roa	28790	0.0367	0.0569	-0.2152	0.013	0.0346	0.0636	0.194
Lev	28790	0.4471	0.1986	0.0638	0.2928	0.4466	0.5978	0.8731
MB	28790	4.0637	2.6449	1.2883	2.4297	3.3367	4.739	17.4268
Growth	28790	0.3949	1.0833	-0.689	-0.0306	0.1323	0.4188	7.8018
Size	28790	22.2616	1.2781	19.8898	21.3587	22.0835	22.9871	26.2096
ACC	28790	0.0693	0.0728	0.0008	0.0219	0.0466	0.0898	0.4018
Top1	28790	0.3404	0.1483	0.0845	0.2243	0.3183	0.4412	0.7402

4. Main empirical results

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4.1. Baseline regression

This paper employs the following regression model to examine the relationship between quantitative mutual funds holdings and stock price crash risk:

$$Crash_{i,t+1} = \beta_0 + \beta_1 QFHPF_{i,t} + \gamma Control_{i,t} + \theta \sum Ind + \delta \sum Year + \varepsilon_{i,t} \quad (5)$$

where *Crash* represents the risk of stock price crash in the next year. To differentiate it from the risk of stock price crash in the current year, *FNCSKEW* or *FDUVOL* is utilized below. *Control* represents a series of control variables, and this regression also includes industry fixed effects and year fixed effects. The results of the baseline regression are shown in Table 2. Whether including partial control variables or including all control variables, the regression coefficient of *QFHPF* is significantly negative at the 1% level, which preliminarily confirms that the higher the proportion of holdings of quantitative mutual funds, the lower the stock price crash risk.

Table 2
Baseline regression.

	(1)	(2)	(3)	(4)
	FNCSKEW	*FDUVOL*	*FNCSKEW*	*FDUVOL*
QFHPF	-0.0604***	-0.0336***	-0.0518***	-0.0311***
	(0.0144)	(0.0091)	(0.0144)	(0.0092)
Dturn	0.0078	0.0054	0.0106*	0.0081**
	(0.0058)	(0.0038)	(0.0058)	(0.0038)
Age	-0.0734***	-0.0565***	-0.0822***	-0.0562***
	(0.0076)	(0.0050)	(0.0080)	(0.0052)
Ret	-22.3239***	-9.8618***	-9.6975**	0.4714
	(4.5779)	(3.0584)	(4.7826)	(3.1937)
NCSKEW	0.0545***	0.0359***	0.0507***	0.0329***
	(0.0062)	(0.0041)	(0.0062)	(0.0041)
Roa	0.6972***	0.3523***	0.5648***	0.3035***
	(0.0762)	(0.0509)	(0.0843)	(0.0557)
Lev			-0.1914***	-0.1015***
			(0.0333)	(0.0219)
MB			0.0198***	0.0115***
			(0.0020)	(0.0014)
Growth			0.0017	0.0014
			(0.0043)	(0.0029)
Size			0.0160***	-0.0036
			(0.0050)	(0.0033)
ACC			0.0698	0.0046
			(0.0586)	(0.0393)
Top1			-0.0980***	-0.0534**
			(0.0320)	(0.0212)
Constant	0.0354	0.0634	-0.2469**	0.1650**

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	(0.0495)	(0.0386)	(0.1136)	(0.0773)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	28790	28790	28790	28790
adj. R^2	0.0519	0.0566	0.0554	0.0602

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.2. Robustness tests

4.2.1. Instrumental variable method

This subsection uses the instrumental variable method to mitigate potential endogeneity issues. We use the average value of *QFHPF* in other companies in the same industry as the instrumental variable. The results in Table 3 show that the coefficient of *QFHPF_IV* in the first stage is significantly positive at the 1% level and the *Cragg-Donald Wald F* statistic is greater than 10, which means *FHPF_IV* is not a weak instrumental variable. In the second stage, the regression coefficients of *QFHPF* are all significantly negative, that is, after addressing the potential endogeneity issues, the main conclusion of this study remain unchanged.

Table 3

Instrumental variable method.

	First stage	Second stage	
	(1)	(2)	(3)
	QFHPF	*FNCSKEW*	*FDUVOL*
QFHPF		-0.8670***	-0.6116***
		(0.3333)	(0.2215)
QFHPF_IV	0.2041***		
	(0.0316)		
Dturn	-0.0043*	0.0070	0.0056
	(0.0025)	(0.0064)	(0.0042)
Age	-0.0164***	-0.0959***	-0.0662***
	(0.0044)	(0.0104)	(0.0069)
Ret	20.4053***	6.4888	12.0551**
	(2.3458)	(8.4732)	(5.6843)
NCSKEW	-0.0004	0.0502***	0.0327***
	(0.0029)	(0.0066)	(0.0044)
Roa	0.3713***	0.8636***	0.5167***
	(0.0418)	(0.1540)	(0.1026)
Lev	0.1308***	-0.0854	-0.0261
	(0.0185)	(0.0573)	(0.0382)
MB	-0.0092***	0.0124***	0.0062**
	(0.0010)	(0.0037)	(0.0025)
Growth	0.0002	0.0018	0.0013
	(0.0022)	(0.0047)	(0.0032)
Size	-0.0293***	-0.0074	-0.0203***

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	(0.0025)	(0.0112)	(0.0074)
ACC	0.0378	0.1049*	0.0292
	(0.0268)	(0.0634)	(0.0427)
Top1	0.0552***	-0.0530	-0.0207
	(0.0172)	(0.0399)	(0.0267)
Constant	0.7389***	0.3573	0.5981***
	(0.0573)	(0.2798)	(0.1869)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Cragg-Donald Wald F statistic	53.501		
N	28724	28724	28724

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.2.2. Heckman two-stage regression

The influence of *QFHPF* on stock price crash risk may be affected by the issue of self-selection bias. To mitigate self-selection bias, Heckman two-stage regression was conducted in this subsection. In the first stage, Probit regression is conducted with a dummy variable indicating whether it is held by quantitative mutual funds (*QFHPF_D*) as the dependent variable. The independent variables include some other variables in addition to the control variables in the baseline regression, and the Inverse Mills Ratio (*IMR*) is calculated accordingly. In the second stage, *IMR* is introduced into the baseline regression as a control variable. Table 4 shows that the coefficients of *IMR* are not significant, suggesting that there is no serious self-selection bias problem. Furthermore, the coefficients of *QFHPF* are significantly negative at the 1% level, indicating that the main findings of this study remain robust after addressing the concern of self-selection bias.

Table 4
Heckman two-stage regression.

	First stage	Second stage	
	(1)	(2)	(3)
	QFHPF_D	*FNCSKEW*	*FDUVOL*
QFHPF		-0.0833***	-0.0444***
		(0.0155)	(0.0102)
Dturn	0.0154	0.0137**	0.0089**
	(0.0149)	(0.0060)	(0.0040)
Age	-0.1555***	-0.0802***	-0.0527***
	(0.0241)	(0.0085)	(0.0056)
Ret	73.7836***	-7.4614	1.7618
	(11.9053)	(5.3962)	(3.5629)
NCSKEW	0.0792***	0.0505***	0.0334***
	(0.0153)	(0.0067)	(0.0044)
Roa	3.3360***	0.6099***	0.3012***
	(0.2142)	(0.1042)	(0.0688)
Lev	-1.5393***	-0.1585***	-0.0855***
	(0.0885)	(0.0374)	(0.0247)

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MB	0.0707***	0.0209***	0.0130***
	(0.0055)	(0.0023)	(0.0015)
Growth	-0.0310***	-0.0002	0.0007
	(0.0097)	(0.0048)	(0.0032)
Size	0.6209***	0.0084	-0.0098**
	(0.0173)	(0.0072)	(0.0048)
ACC	0.0234	0.0330	-0.0360
	(0.1491)	(0.0653)	(0.0431)
Top1	-0.6223***	-0.0907***	-0.0470**
	(0.0950)	(0.0324)	(0.0214)
MER	-0.4720***		
	(0.1778)		
EqN	0.1596***		
	(0.0274)		
Inv	0.9641***		
	(0.0761)		
Wage	0.2458***		
	(0.0196)		
MH	1.1642***		
	(0.1016)		
IMR		0.0208	-0.0062
		(0.0356)	(0.0235)
Constant	-16.4792***	-0.0155	0.3262***
	(0.4049)	(0.1811)	(0.1196)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	28790	28790	28790
Pseudo R2/Wald chi2	0.3090	1587.57	1668.12

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The detailed definitions of control variables, including management expense rate (*MER*), nature of property rights (*EqN*), institutional investors' shareholding ratio (*Inv*), management compensation (*Wage*), and management shareholding ratio (*MH*), are provided in the appendix.

4.2.3. Alternative measures of stock price crash risk

Considering the significant differences among various sub-markets (Main-Board Market, Growth Enterprises Market, and STAR Market) in the A-share market, this subsection replaces the dependent variable with *FNCSKEW* and *FDUVOL*, both calculated using the value-weighted average return of each sub-market (*FNCSKEW_d* and *FDUVOL_d*). The results in Table 5 indicate that the baseline finding remains robust.

Table 5

Alternative measures of stock price crash risk.

	(1)	(2)
	FNCSKEW_d	*FDUVOL_d*

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QFHPF	-0.0500***	-0.0284***
	(0.0146)	(0.0094)
Dturn	0.0006	-0.0014
	(0.0059)	(0.0039)
Age	-0.0337***	-0.0123**
	(0.0083)	(0.0055)
Ret	-4.4501	3.8314
	(4.8982)	(3.2416)
NCSKEW	0.0391***	0.0243***
	(0.0063)	(0.0041)
Roa	0.5896***	0.3130***
	(0.0859)	(0.0572)
Lev	-0.1617***	-0.0794***
	(0.0348)	(0.0234)
MB	0.0218***	0.0134***
	(0.0021)	(0.0014)
Growth	0.0037	0.0033
	(0.0044)	(0.0029)
Size	0.0262***	0.0054
	(0.0052)	(0.0034)
ACC	0.0684	-0.0063
	(0.0599)	(0.0398)
Top1	-0.0619*	-0.0297
	(0.0340)	(0.0230)
Constant	-0.5982***	-0.1358*
	(0.1192)	(0.0775)
Industry FE	Yes	Yes
Year FE	Yes	Yes
N	28790	28790
adj. R^2	0.0450	0.0470

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The main difference between *FNCSKEW_d* and *FNCSKEW* lies in the calculation of the firm-specific weekly return of stock *i* in week *j* ($w_{i,j}$). When calculating *FNCSKEW_d*, if stock *i* belongs to the Growth Enterprises Market, then $r_{m,j}$ in Eq. (1) represents the value-weighted average return of the Growth Enterprises Market for week *j*. If stock *i* belongs to the STAR Market, $r_{m,j}$ in Eq. (1) represents the value-weighted average return of the STAR Market for week *j*. When calculating *FNCSKEW*, $r_{m,j}$ in Eq. (1) represents the value-weighted average return of the A-share market for week *j*. The difference between *FDUVOL_d* and *FDUVOL* is similar.

4.2.4. Exclude the impact of mutual funds

In order to verify that the inhibitory effect of quantitative mutual funds holdings (*QFHPF*) on stock price crash risk is not caused by its institutional investor nature, this subsection tests the relationship between mutual funds holdings (*FHP*) and stock price crash risk. Table 6 shows that the coefficients of *FHP* are significantly positive at the 1% level, indicating that mutual funds

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holdings (*FHP*) may exacerbate stock price crash risk due to herding effect. This also implies that the inhibitory effect of quantitative mutual funds holdings (*QFHPF*) on stock price crash risk is not due to its institutional investor nature.

Table 6

mutual funds holdings (*FHP*) and stock price crash risk.

	(1)	(2)
	FNCSKEW	*FDUVOL*
FHP	1.1190***	0.7086***
	(0.0661)	(0.0470)
Dturn	0.0104*	0.0080**
	(0.0058)	(0.0038)
Age	-0.0722***	-0.0499***
	(0.0079)	(0.0051)
Ret	-7.3663	1.9821
	(4.7588)	(3.1761)
NCSKEW	0.0421***	0.0275***
	(0.0062)	(0.0041)
Roa	0.2358***	0.0957*
	(0.0868)	(0.0572)
Lev	-0.1390***	-0.0681***
	(0.0330)	(0.0218)
MB	0.0127***	0.0071***
	(0.0021)	(0.0014)
Growth	0.0014	0.0012
	(0.0043)	(0.0028)
Size	-0.0037	-0.0161***
	(0.0051)	(0.0034)
ACC	0.0450	-0.0111
	(0.0587)	(0.0394)
Top1	-0.0410	-0.0171
	(0.0320)	(0.0211)
Constant	0.0809	0.3739***
	(0.1152)	(0.0781)
Industry FE	Yes	Yes
Year FE	Yes	Yes
N	28790	28790
adj. R^2	0.0621	0.0664

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The detailed definition of *FHP* is provided in the appendix.

4.2.5. Propensity Score Matching

Compared to stocks not held by quantitative mutual funds, stocks held by quantitative mutual funds may possess certain different characteristics, which could potentially impact stock price

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stability. This subsection employs the propensity score matching (PSM) method to alleviate such endogeneity problems. First, the Logit model is estimated, where the dependent variable is the dummy variable *QFHPF_D*, and the independent variables include all control variables from the baseline regression. Second, the one-to-one nearest neighbor matching method is used to generate new samples, and subsequently, the baseline regression model is re-executed. The results in Table 7 show that the main findings of this paper still hold after eliminating the systematic differences between the treatment group and the control group.

Table 7
Propensity Score Matching.

	(3)	(4)
	FNCSKEW	*FDUVOL*
QFHPF	-0.0511***	-0.0312***
	(0.0145)	(0.0092)
Dturn	0.0095	0.0074*
	(0.0060)	(0.0039)
Age	-0.0819***	-0.0561***
	(0.0082)	(0.0053)
Ret	-10.0690**	0.1545
	(4.8441)	(3.2239)
NCSKEW	0.0502***	0.0324***
	(0.0063)	(0.0041)
Roa	0.5544***	0.2970***
	(0.0857)	(0.0565)
Lev	-0.1868***	-0.0939***
	(0.0344)	(0.0225)
MB	0.0194***	0.0108***
	(0.0021)	(0.0014)
Growth	0.0012	0.0009
	(0.0043)	(0.0029)
Size	0.0152***	-0.0046
	(0.0056)	(0.0036)
ACC	0.0779	0.0120
	(0.0599)	(0.0400)
Top1	-0.1039***	-0.0552**
	(0.0327)	(0.0214)
Constant	-0.2243*	0.1885**
	(0.1235)	(0.0820)
Industry FE	Yes	Yes
Year FE	Yes	Yes
N	28072	28072
adj. R^2	0.0550	0.0597

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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4.3. Cross-sectional heterogeneity

4.3.1. Management characteristics

Some special experiences of the senior management of listed companies may affect the stock price crash risk (Cui et al., 2022; Wang et al., 2022), so it is interesting to examine whether the impact of quantitative mutual funds holdings on the stock price crash risk is different among listed companies with different senior management backgrounds. In this subsection, according to whether the senior management of listed companies has overseas background, the sample data is divided into two groups (MOB and Non-MOB) for regression, and the results are shown in Table 8. Thanks to the powerful information search and information processing capabilities of quantitative investment technology, it can effectively identify some factors conducive to stock price stability in listed companies and better exert their effectiveness, making the inhibitory effect of quantitative mutual funds holdings on stock price crash risk more significant in the MOB subsample.

Table 8

Management overseas background.

	FNCSKEW		*FDUVOL*	
	(1)	(2)	(3)	(4)
	Non-MOB	MOB	Non-MOB	MOB
QFHPF	-0.0203	-0.0826***	-0.0124	-0.0489***
	(0.0207)	(0.0198)	(0.0132)	(0.0127)
Dturn	0.0128	0.0090	0.0082	0.0081
	(0.0090)	(0.0078)	(0.0058)	(0.0051)
Age	-0.0829***	-0.0822***	-0.0556***	-0.0569***
	(0.0120)	(0.0109)	(0.0078)	(0.0071)
Ret	-5.9905	-11.1985*	4.5702	-2.1177
	(7.2017)	(6.3729)	(4.7420)	(4.3444)
NCSKEW	0.0556***	0.0413***	0.0329***	0.0299***
	(0.0092)	(0.0085)	(0.0060)	(0.0056)
Roa	0.5752***	0.5429***	0.2791***	0.3126***
	(0.1286)	(0.1109)	(0.0844)	(0.0735)
Lev	-0.1890***	-0.1977***	-0.1067***	-0.1001***
	(0.0500)	(0.0438)	(0.0333)	(0.0290)
MB	0.0168***	0.0233***	0.0101***	0.0134***
	(0.0031)	(0.0027)	(0.0021)	(0.0018)
Growth	-0.0001	0.0039	-0.0009	0.0034
	(0.0059)	(0.0065)	(0.0040)	(0.0042)
Size	0.0124	0.0178***	-0.0062	-0.0027
	(0.0079)	(0.0065)	(0.0053)	(0.0043)
ACC	-0.0093	0.1130	-0.0539	0.0411
	(0.0854)	(0.0809)	(0.0571)	(0.0544)
Top1	-0.0737	-0.1181***	-0.0483	-0.0542*
	(0.0471)	(0.0423)	(0.0307)	(0.0285)
Constant	-0.1516	-0.3303**	0.2191*	0.1250

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	(0.1763)	(0.1502)	(0.1187)	(0.1027)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	13067	15723	13067	15723
adj. R^2	0.0621	0.0522	0.0677	0.0563
Subsample comparison of coefficients on QFHPF	Chi-Square = 4.83 P-value = 0.0280		Chi-Square = 4.09 P-value = 0.0431	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3.2. Internal governance efficiency

Generally speaking, companies with higher internal governance efficiency can better restrain the self-interest behavior of management, allowing their disclosed information to objectively reflect the true operating condition of the company. High-quality information sources provide conditions for quantitative investment techniques to fully exert their effectiveness. Therefore, we predict that the negative relationship between quantitative mutual funds holdings and stock price crash risk is more significant in the subsample with higher internal governance efficiency. In Table 9, this subsection divides the enterprises whose CEO does not also serve as Chairman into the subsample with high internal governance efficiency and the enterprises whose CEO also serves as Chairman into the subsample with poor internal governance efficiency. The regression results are consistent with our prediction.

Table 9

CEO-Chairman Duality.

	FNCSKEW		*FDUVOL*	
	(1) NOT-DUAL	(2) DUAL	(3) NOT-DUAL	(4) DUAL
QFHPF	-0.0708***	0.0037	-0.0433***	0.0055
	(0.0166)	(0.0297)	(0.0106)	(0.0186)
Dturn	0.0116*	0.0058	0.0066	0.0111
	(0.0067)	(0.0122)	(0.0044)	(0.0078)
Age	-0.0813***	-0.0727***	-0.0572***	-0.0522***
	(0.0093)	(0.0173)	(0.0061)	(0.0111)
Ret	-8.4243	-12.3210	0.0112	1.3813
	(5.5811)	(9.5886)	(3.6908)	(6.3575)
NCSKEW	0.0517***	0.0409***	0.0321***	0.0319***
	(0.0070)	(0.0135)	(0.0047)	(0.0087)
Roa	0.5035***	0.7302***	0.2914***	0.3554***
	(0.0988)	(0.1696)	(0.0657)	(0.1102)
Lev	-0.1865***	-0.1999***	-0.1010***	-0.0982**
	(0.0385)	(0.0679)	(0.0254)	(0.0442)
MB	0.0191***	0.0221***	0.0115***	0.0120***
	(0.0024)	(0.0040)	(0.0016)	(0.0026)
Growth	0.0014	0.0037	0.0012	0.0018
	(0.0047)	(0.0100)	(0.0033)	(0.0063)

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Size	0.0140**	0.0248**	-0.0040	-0.0007
	(0.0056)	(0.0111)	(0.0038)	(0.0068)
ACC	0.0417	0.1646	-0.0165	0.0648
	(0.0671)	(0.1215)	(0.0454)	(0.0809)
Top1	-0.0901**	-0.1149*	-0.0419*	-0.0787*
	(0.0367)	(0.0660)	(0.0243)	(0.0436)
Constant	-0.2240*	-0.2183	0.1610*	0.2253
	(0.1259)	(0.2657)	(0.0870)	(0.1570)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	21794	6996	21794	6996
adj. R^2	0.0580	0.0414	0.0635	0.0453
Subsample comparison of coefficients on QFHPF	Chi-Square = 5.15 P-value = 0.0232		Chi-Square = 5.53 P-value = 0.0187	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3.3. Information environment

A favorable external information environment helps investors quickly extract valuable information from the noisy capital market, thereby alleviating information asymmetry and reducing the risk of stock price crash. Financial analysts' professional interpretation of the public information of listed companies is conducive to high-quality information dissemination, thus optimizing the market information environment. Therefore, more analyst coverage also means a better information environment. We divide the sample into two sub-samples according to the level of analyst coverage. Table 10 shows that the inhibitory effect of quantitative mutual funds holdings on stock price crash risk is more significant in the subsample with high analyst coverage. This suggests that a good information environment helps quantitative investment to conduct efficient data mining, which in turn promotes stock price stability.

Table 10

Analyst Coverage.

	FNCSKEW		*FDUVOL*	
	(1)	(2)	(3)	(4)
	Low	High	Low	High
QFHPF	-0.0298*	-0.1159***	-0.0218**	-0.0621***
	(0.0173)	(0.0261)	(0.0107)	(0.0175)
Dturn	0.0052	0.0072	0.0083	0.0018
	(0.0083)	(0.0081)	(0.0053)	(0.0055)
Age	-0.0663***	-0.0625***	-0.0490***	-0.0411***
	(0.0116)	(0.0114)	(0.0073)	(0.0079)
Ret	-12.2330*	-9.5056	-1.1676	1.2689
	(6.3824)	(7.1193)	(4.1280)	(4.9935)
NCSKEW	0.0502***	0.0334***	0.0316***	0.0229***
	(0.0082)	(0.0093)	(0.0053)	(0.0063)
Roa	0.3141***	0.3316**	0.1891***	0.1032

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	(0.1136)	(0.1451)	(0.0731)	(0.0975)
Lev	-0.0765	-0.2301***	-0.0095	-0.1613***
	(0.0467)	(0.0495)	(0.0297)	(0.0346)
MB	0.0097***	0.0250***	0.0036*	0.0182***
	(0.0031)	(0.0030)	(0.0020)	(0.0021)
Growth	0.0068	-0.0055	0.0047	-0.0040
	(0.0054)	(0.0073)	(0.0034)	(0.0052)
Size	-0.0224**	0.0047	-0.0289***	-0.0072
	(0.0089)	(0.0072)	(0.0056)	(0.0049)
ACC	0.0346	0.0502	0.0058	-0.0338
	(0.0801)	(0.0859)	(0.0524)	(0.0606)
Top1	-0.0865*	-0.0874**	-0.0483*	-0.0473
	(0.0454)	(0.0442)	(0.0289)	(0.0309)
Constant	0.4888**	0.0349	0.6467***	0.2717**
	(0.1988)	(0.1651)	(0.1224)	(0.1207)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	15661	13129	15661	13129
adj. R^2	0.0471	0.0695	0.0558	0.0751
Subsample comparison of	Chi-Square = 7.75		Chi-Square = 4.01	
coefficients on QFHPF	P-value = 0.0054		P-value = 0.0452	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Analyst Coverage is measured by the natural logarithm of the number of analysts (teams) that have followed the listed company each year. This table is grouped based on the annual industry median of Analyst Coverage, with samples exceeding the annual industry median categorized as the High Analyst Coverage group, and the remaining samples categorized as the Low Analyst Coverage group.

4.3.4. Profit Volatility

Generally speaking, companies with stable operating profits tend to have relatively stable stock prices, while companies with larger fluctuations in operating profits are more likely to magnify investors' greed or panic, prompting them to make irrational investment decisions and leading to sharp increases or declines in stock prices. Quantitative investment is more rational than discretionary investment, so it is reasonable to believe that the negative effect of quantitative mutual funds holdings on stock price crash risk is stronger in companies with high profit volatility. In view of this, this subsection divides the sample into two subsamples for verification according to the profit volatility, and the results in Table 11 are consistent with the above reasoning.

Table 11
Profit Volatility.

	FNCSKEW		*FDUVOL*	
	(1)	(2)	(3)	(4)
	Low	High	Low	High
QFHPF	-0.0188	-0.0826***	-0.0107	-0.0495***
	(0.0203)	(0.0203)	(0.0130)	(0.0129)

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Dturn	0.0128	0.0082	0.0116**	0.0043
	(0.0079)	(0.0085)	(0.0052)	(0.0056)
Age	-0.0788***	-0.0827***	-0.0552***	-0.0558***
	(0.0111)	(0.0112)	(0.0072)	(0.0073)
Ret	-7.1444	-11.7798*	2.1113	-1.9014
	(7.3038)	(6.4233)	(4.7952)	(4.3069)
NCSKEW	0.0523***	0.0466***	0.0349***	0.0301***
	(0.0086)	(0.0090)	(0.0057)	(0.0058)
Roa	0.9245***	0.4997***	0.4541***	0.2584***
	(0.1989)	(0.0969)	(0.1320)	(0.0645)
Lev	-0.2011***	-0.1619***	-0.1200***	-0.0834***
	(0.0493)	(0.0460)	(0.0334)	(0.0300)
MB	0.0200***	0.0187***	0.0114***	0.0114***
	(0.0033)	(0.0026)	(0.0022)	(0.0017)
Growth	-0.0058	0.0070	-0.0041	0.0051
	(0.0069)	(0.0055)	(0.0044)	(0.0037)
Size	0.0147**	0.0168**	-0.0035	-0.0033
	(0.0070)	(0.0069)	(0.0046)	(0.0046)
ACC	0.0952	0.0333	0.0110	0.0029
	(0.0953)	(0.0749)	(0.0643)	(0.0502)
Top1	-0.0703	-0.1313***	-0.0395	-0.0703**
	(0.0441)	(0.0440)	(0.0293)	(0.0290)
Constant	-0.2919*	-0.1975	0.1712	0.1456
	(0.1593)	(0.1598)	(0.1060)	(0.1108)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	14657	14133	14657	14133
adj. R^2	0.0548	0.0564	0.0617	0.0591
Subsample comparison of coefficients on QFHPF	Chi-Square = 5.06 P-value = 0.0245		Chi-Square = 4.62 P-value = 0.0316	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Profit Volatility is measured by the standard deviation of the ratio of *EBIT* (Earnings Before Interest and Taxes) to total assets of the listed company over the past three years. This table is grouped based on the annual industry median of Profit Volatility, with samples exceeding the annual industry median categorized as the High Profit Volatility group, and the remaining samples categorized as the Low Profit Volatility group.

4.4. Further analysis

Studying the mechanism by which quantitative mutual funds holdings affect the risk of stock price crash is of great importance. On the one hand, quantitative investment technology possesses extraordinary capabilities in information collection, conversion, and processing. This enables stock prices to more promptly reflect specific market information, thereby allowing quantitative mutual funds holdings to mitigate stock price crash risk by reducing volatility. On the other hand,

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quantitative investment technology can achieve more accurate assessment of stock prices by introducing cutting-edge algorithms. Meanwhile, its more rational investment style compared with discretionary investment can also help avoid speculative bubbles caused by excessive greed. Therefore, we infer that quantitative mutual funds holdings can restrain stock price crash risk by stabilizing stock valuation. In this subsection, stock return volatility (*Sigma*) and stock valuation (*TobinQ*) are employed as mediating variables and the following three regressions are used to examine the possible mechanisms.

$$Crash_{i,t+1} = \beta_0 + \beta_1 QFHPF_{i,t} + \gamma Control_{i,t} + \theta \sum Ind + \delta \sum Year + \varepsilon_{i,t} \quad (6)$$

$$Mediator_{i,t} = \alpha_0 + \alpha_1 QFHPF_{i,t} + \gamma Control_{i,t} + \theta \sum Ind + \delta \sum Year + \tau_{i,t} \quad (7)$$

$$Crash_{i,t+1} = \beta'_0 + \beta'_1 Mediator_{i,t} + \beta'_2 QFHPF_{i,t} + \gamma Control_{i,t} + \theta \sum Ind + \delta \sum Year + \omega_{i,t} \quad (8)$$

Table 12 shows that higher *QFHPF* is associated with lower *Sigma*, which means lower risk of stock price crash, and the proportion of total effect that is mediated by *Sigma* is 6.4% and 5.2%. Similarly, Table 13 shows that higher *QFHPF* is associated with lower *TobinQ*, which indicates lower stock price crash risk, and the proportion of total effect that is mediated by *TobinQ* is 3.7% and 2.7%.

Table 12

Possible mechanism: *Sigma*.

	(1)	(2)	(3)	(4)	(5)
	FNCSKEW	*Sigma*	*FNCSKEW*	*FDUVOL*	*FDUVOL*
QFHPF	-0.0518***	-0.0005***	-0.0485***	-0.0311***	-0.0295***
	(0.0144)	(0.0001)	(0.0144)	(0.0092)	(0.0092)
Sigma			6.3087***		3.0857***
			(1.0655)		(0.6930)
Dturn	0.0106*	0.0014***	0.0018	0.0081**	0.0038
	(0.0058)	(0.0000)	(0.0060)	(0.0038)	(0.0039)
Age	-0.0822***	-0.0008***	-0.0769***	-0.0562***	-0.0536***
	(0.0080)	(0.0001)	(0.0081)	(0.0052)	(0.0053)
Ret	-9.6975**	-15.4346***	87.6740***	0.4714	48.0984***
	(4.7826)	(0.0573)	(16.8445)	(3.1937)	(11.0532)
NCSKEW	0.0507***	0.0003***	0.0490***	0.0329***	0.0321***
	(0.0062)	(0.0000)	(0.0062)	(0.0041)	(0.0041)
Roa	0.5648***	0.0021***	0.5517***	0.3035***	0.2971***
	(0.0843)	(0.0005)	(0.0842)	(0.0557)	(0.0558)
Lev	-0.1914***	0.0019***	-0.2032***	-0.1015***	-0.1073***
	(0.0333)	(0.0002)	(0.0333)	(0.0219)	(0.0220)
MB	0.0198***	0.0000***	0.0195***	0.0115***	0.0114***
	(0.0020)	(0.0000)	(0.0020)	(0.0014)	(0.0014)
Growth	0.0017	0.0000	0.0017	0.0014	0.0014
	(0.0043)	(0.0000)	(0.0043)	(0.0029)	(0.0029)
Size	0.0160***	-0.0007***	0.0205***	-0.0036	-0.0014

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	(0.0050)	(0.0000)	(0.0050)	(0.0033)	(0.0033)
ACC	0.0698	0.0022***	0.0559	0.0046	-0.0022
	(0.0586)	(0.0003)	(0.0586)	(0.0393)	(0.0393)
Top1	-0.0980***	-0.0003	-0.0963***	-0.0534**	-0.0525**
	(0.0320)	(0.0002)	(0.0320)	(0.0212)	(0.0211)
Constant	-0.2469**	0.0467***	-0.5417***	0.1650**	0.0208
	(0.1136)	(0.0008)	(0.1219)	(0.0773)	(0.0830)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
N	28790	28790	28790	28790	28790
adj. R^2	0.0554	0.9568	0.0566	0.0602	0.0609
Sobel Test		-4.580***		-3.799***	
Percentage		6.4%		5.2%	

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The detailed definition of *Sigma* is provided in the appendix.

Table 13

Possible mechanism: *TobinQ*.

	(1) *FNCSKEW*	(2) *TobinQ*	(3) *FNCSKEW*	(4) *FDUVOL*	(5) *FDUVOL*
QFHPF	-0.0518***	-0.0676***	-0.0499***	-0.0311***	-0.0303***
	(0.0144)	(0.0135)	(0.0144)	(0.0092)	(0.0092)
TobinQ			0.0280***		0.0125**
			(0.0077)		(0.0051)
Dturn	0.0106*	-0.0267***	0.0114*	0.0081**	0.0084**
	(0.0058)	(0.0042)	(0.0058)	(0.0038)	(0.0038)
Age	-0.0822***	0.0560***	-0.0838***	-0.0562***	-0.0569***
	(0.0080)	(0.0100)	(0.0080)	(0.0052)	(0.0052)
Ret	-9.6975**	-66.2930***	-7.8382	0.4714	1.3019
	(4.7826)	(6.3502)	(4.8266)	(3.1937)	(3.2218)
NCSKEW	0.0507***	0.0158***	0.0502***	0.0329***	0.0327***
	(0.0062)	(0.0053)	(0.0062)	(0.0041)	(0.0041)
Roa	0.5648***	2.2370***	0.5020***	0.3035***	0.2754***
	(0.0843)	(0.1327)	(0.0870)	(0.0557)	(0.0572)
Lev	-0.1914***	-3.6473***	-0.0891**	-0.1015***	-0.0558*
	(0.0333)	(0.0622)	(0.0437)	(0.0219)	(0.0286)
MB	0.0198***	0.4058***	0.0084**	0.0115***	0.0065***
	(0.0020)	(0.0085)	(0.0038)	(0.0014)	(0.0025)
Growth	0.0017	-0.0055	0.0019	0.0014	0.0015
	(0.0043)	(0.0051)	(0.0043)	(0.0029)	(0.0029)
Size	0.0160***	-0.0431***	0.0172***	-0.0036	-0.0031
	(0.0050)	(0.0083)	(0.0050)	(0.0033)	(0.0033)
ACC	0.0698	0.0181	0.0693	0.0046	0.0043
	(0.0586)	(0.0662)	(0.0586)	(0.0393)	(0.0393)

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Top1	-0.0980***	-0.1146***	-0.0948***	-0.0534**	-0.0519**
	(0.0320)	(0.0422)	(0.0322)	(0.0212)	(0.0212)
Constant	-0.2469**	2.8502***	-0.3269***	0.1650**	0.1293
	(0.1136)	(0.1961)	(0.1157)	(0.0773)	(0.0786)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
N	28790	28790	28790	28790	28790
adj. R^2	0.0554	0.8147	0.0559	0.0602	0.0604
Sobel Test		-3.283***			-2.413**
Percentage		3.7%			2.7%

Notes: Robust standard errors clustered by firm in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The detailed definition of *TobinQ* is provided in the appendix.

5. Conclusion

Different from the limited computing power and attention of the human brain, the computing power of quantitative investment strategies is approaching infinity with the continuous improvement of computer hardware and the rapid development of digital technology. This enables it to collect and process noisy capital market information more efficiently. Furthermore, quantitative investment, as specific computer programs developed by incorporating previous investment experience, possess the capability to make optimal trading decisions that align with the risk-return targets of the strategies in complex information environments, without being swayed by investors' subjective preferences. This renders its investment more rational and effectively prevents speculative bubbles caused by excessive greed. From the perspective of quantitative mutual funds holdings, this study finds that the widespread application of quantitative investment technology can suppress stock price crash risk by smoothing the volatility of stock returns and optimizing the valuation of firms. Moreover, this effect is more pronounced in subsamples characterized by executives with overseas backgrounds, higher internal governance efficiency, greater analyst attention, and higher profit volatility. Our findings confirm the positive role of quantitative investment in stabilizing stock prices, providing evidence supporting the widespread application of financial technology and cutting-edge algorithms in the investment community.

Declaration of Competing Interest

None

Appendix. Definitions of variables

NCSKEW	The negative skewness of firm-specific weekly returns over the fiscal year.
DUVOL	The log of the ratio of the standard deviations of down-week to up-week firm-specific weekly returns.
QFHPF	The ratio of shares held by quantitative mutual funds to the total shares held by all mutual funds.

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Dturn	The difference between the stock turnover rate in year t and the stock turnover rate in year t-1, divided by the stock turnover rate in year t.
Age	The natural logarithm of the number of years a firm has been listed plus one.
Ret	Mean of the stock's firm-specific weekly return.
Roa	Returns on total assets.
Lev	Total debts divided by total assets.
MB	The market value of equity divided by the book value of equity.
Growth	The difference between the current year's operating income and the prior year's operating income, divided by the prior year's operating income.
Size	The natural logarithm of market capitalization.
ACC	The absolute value of discretionary accruals, where discretionary accruals are estimated from the modified Jones model.
Top1	Shareholding ratio of the largest shareholder.
QFHPF_D	A dummy variable takes a value of 1 when listed company is held by quantitative mutual funds, and 0 otherwise.
MER	Management expenses over total assets.
EqN	A dummy variable, where the value is 1 for state-owned enterprises and 0 for non-state-owned enterprises.
Inv	The proportion of shares of listed companies held by institutional investors.
Wage	The total compensation of top three senior executives in listed companies.
MH	The ratio of the number of shares held by senior management to the total number of shares in the listed company.
FHP	The ratio of the number of shares held by mutual funds to the total number of shares of listed companies.
Sigma	Standard deviation of the stock's firm-specific weekly return.
TobinQ	Total market value divided by total assets.

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